a11y

(Accessibility)



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- seit 2017 bei Coop in der internen Entwicklung
- 2019 – 2023 Bachelor an der FHNW
- 2025 – 2027 MSE (Master of Science in Engineering)

Stimmungsbild & Vorwissen



Arten von Einschränkungen

Permanent



One arm

Temporary



Broken arm

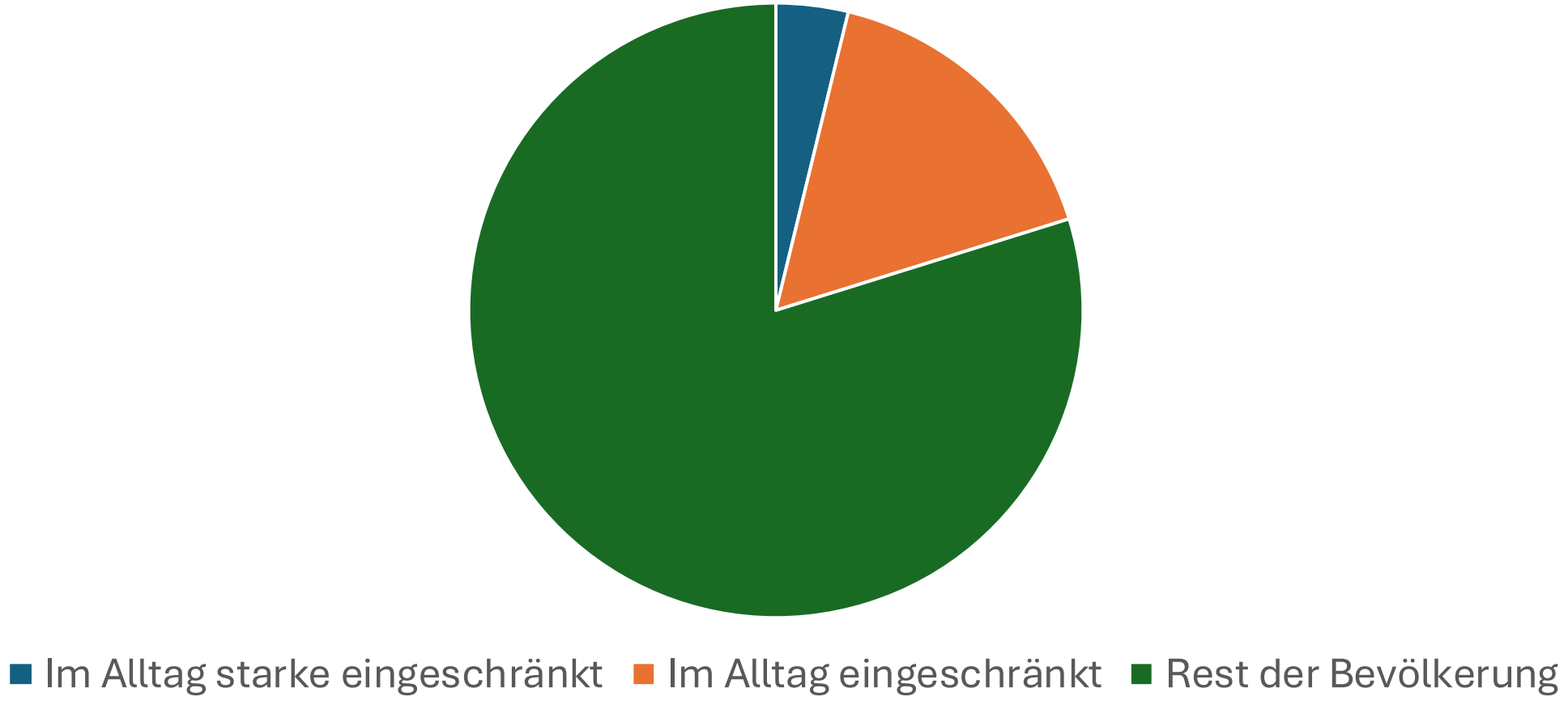
Situational



Holding a baby

https://uxdesign.cc/understanding-accessibility-makes-you-a-better-insert-your-name-here-1f7a15c50e7f

Schweizer Bevölkerung – 8’962’300 (permanente Beeinträchtigungen - Stand 2023)



https://www.bfs.admin.ch/bfs/de/home/statistiken/wirtschaftliche-soziale-situation-bevoelkerung/gleichstellung-menschen-behinderungen/behinderungen.html



Web-Applikationen für Accessibility (a11y) optimieren

Web-Applikationen optimieren, um 1ms schneller zu sein

https://en.meming.world/images/en/0/07/Drakeposting.jpg

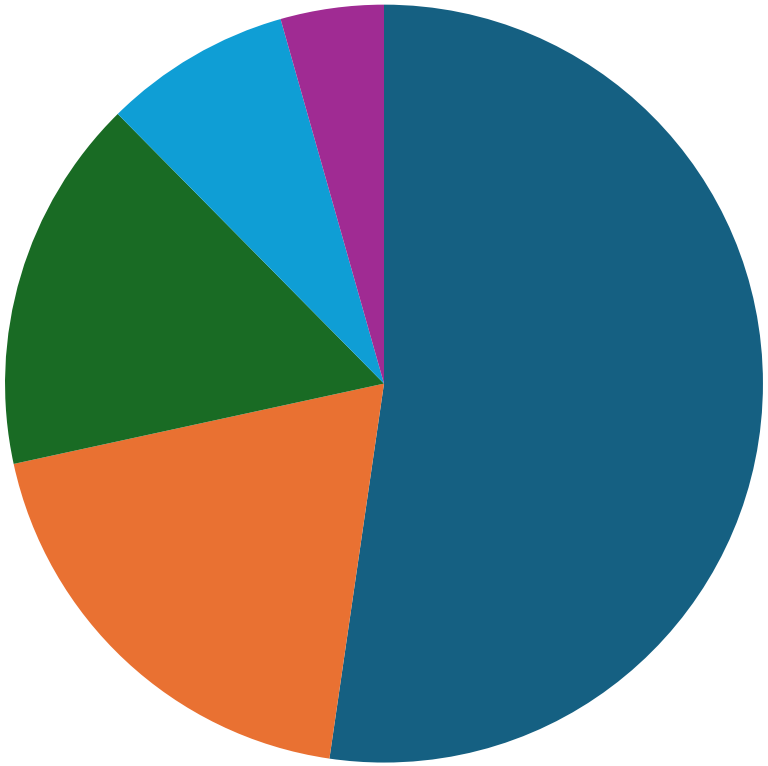
Jenseits von Maus und Bildschirm





https://i0.wp.com/css-tricks.com/wp-content/uploads/2022/08/jaws-nvda-voiceover.jpg
https://1lowvision.com/image/catalog/pacificvision/HIMS/BrailleEMotion/AngledRight-2-e1709748782159.png
https://resource.logitech.com/c_fill,q_auto,f_auto,dpr_1.0/d_transparent.gif/content/dam/logitech/en/products/keyboards/k250/gallery/delorean-gallery-images/deu/k-250-graphite-top-view-2-deu.png

Verwendete Browser in Kombination mit Screenreadern



Chrome 52.3% Microsoft Edge 19.3% Firefox 16% Safari 8% restliche Browser 4.4%

https://webaim.org/projects/screenreadersurvey10/

Semantik

Banane =



https://pngimg.com/uploads/banana/banana_PNG835.png

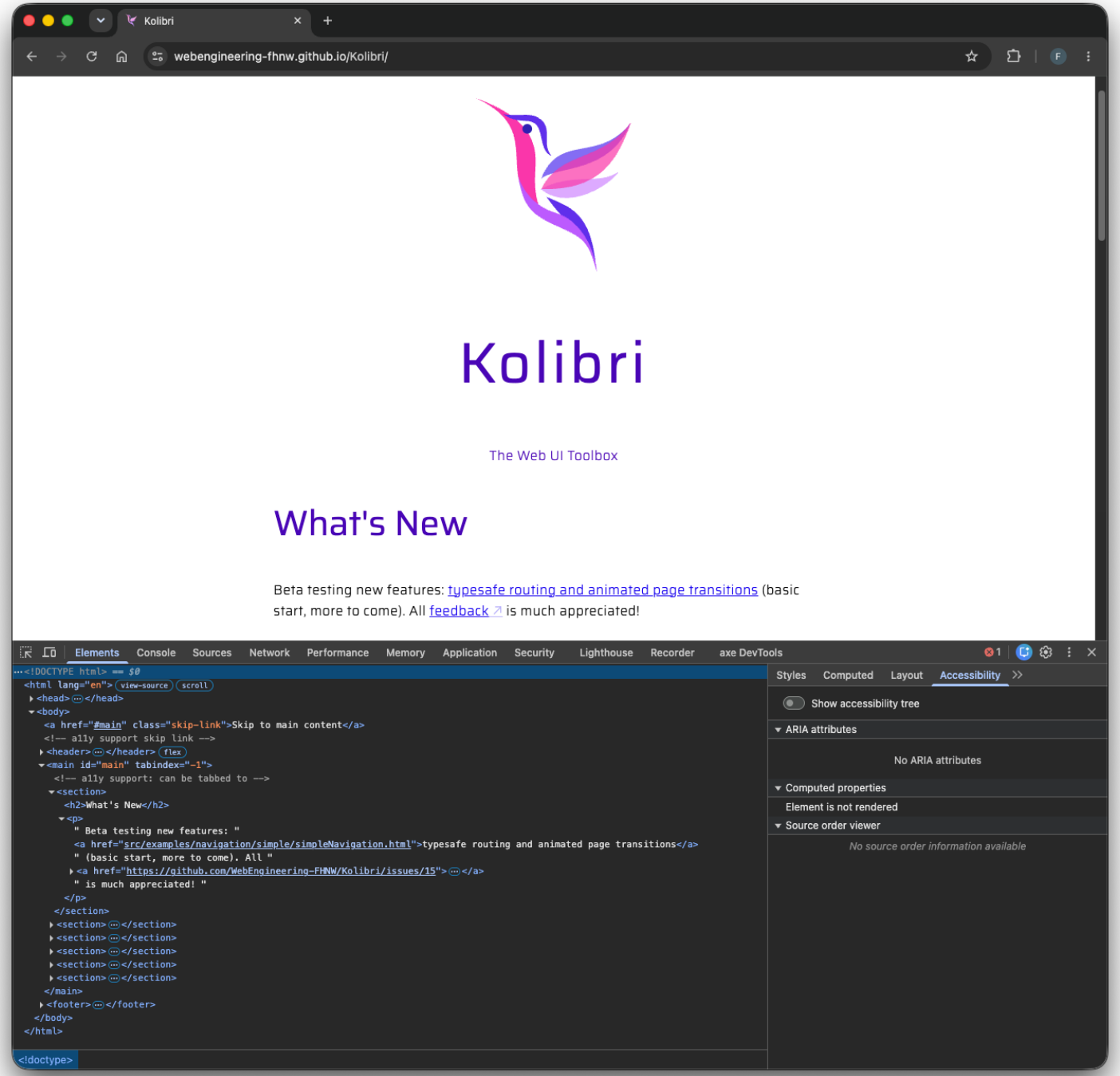
```
<button>Button Text</button>
```



vs

```
<div>Button Text</div>
```

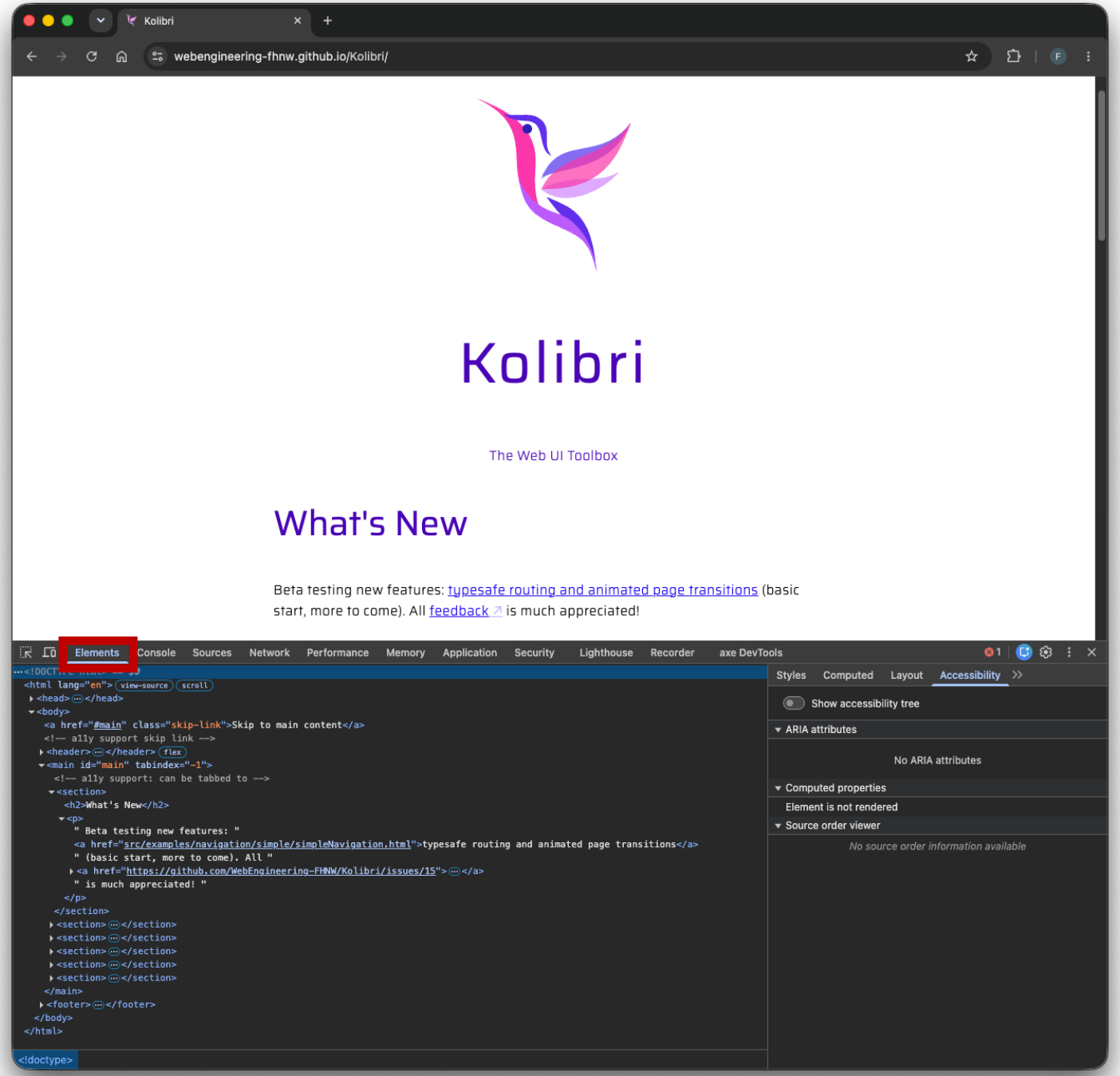
Button Text


Kolibri
webengineering-fhnw.github.io/Kolibri/
Kolibri
The Web UI Toolbox
What's New
Beta testing new features: typesafe routing and animated page transitions (basic start, more to come). All feedback is much appreciated!
Elements Console Sources Network Performance Memory Application Security Lighthouse Recorder axe DevTools
<!DOCTYPE html> == $0
<html lang="en"> view-source scroll
<head> </head>
<body>
Skip to main content
<!-- a11y support skip link -->
<header> </header> flex
<main id="main" tabindex="-1">
<!-- a11y support: can be tabbed to -->
<section>
<h2>What's New</h2>

" Beta testing new features: "
typesafe routing and animated page transitions
" (basic start, more to come). All "

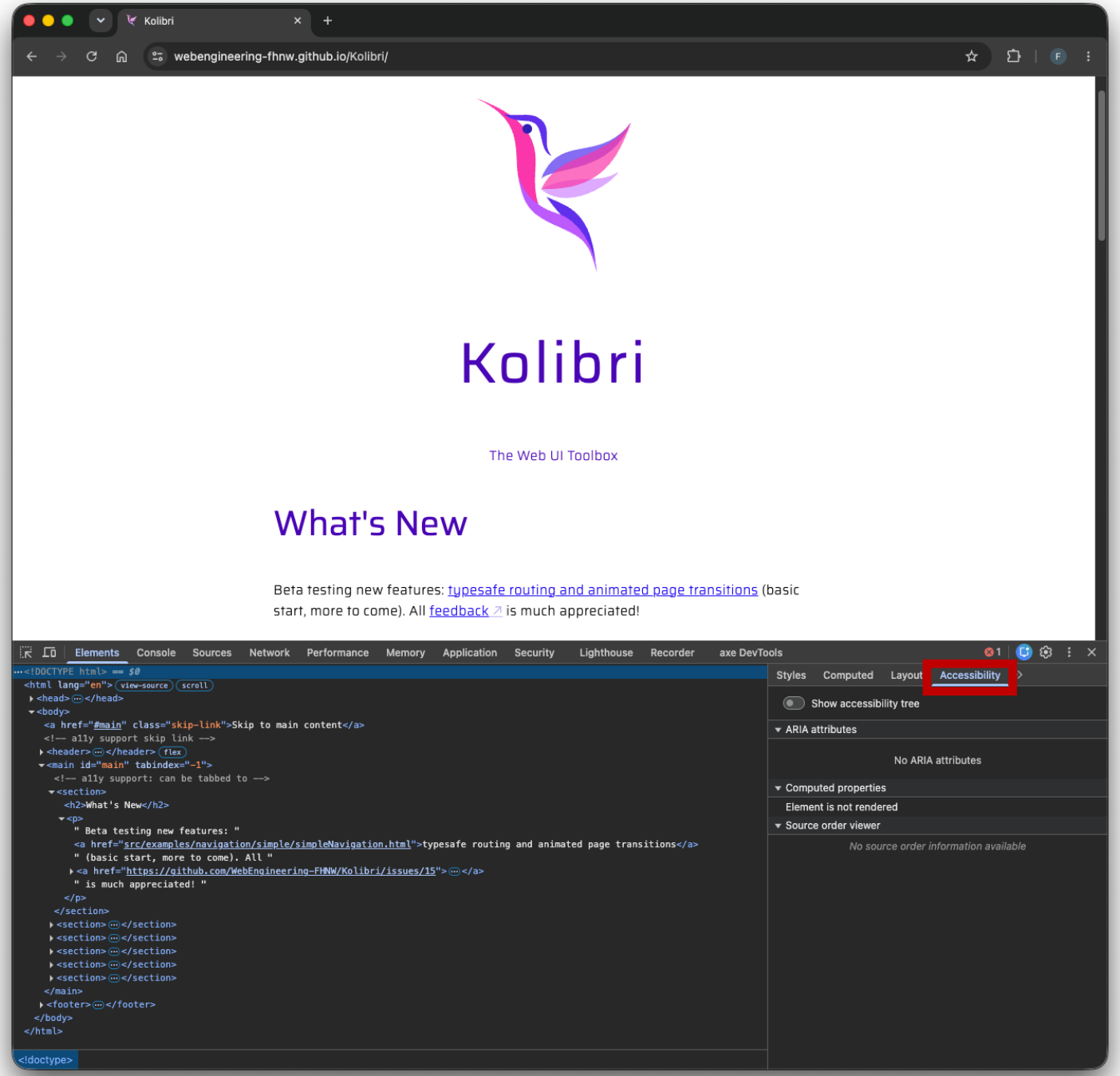
" is much appreciated! "

</section>
<section> </section>
<section> </section>
<section> </section>
<section> </section>
<section> </section>
</main>
<footer> </footer>
</body>
</html>
<!doctype>
Styles Computed Layout Accessibility
Show accessibility tree
ARIA attributes
No ARIA attributes
Computed properties
Element is not rendered
Source order viewer
No source order information available


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<a href="https://github.com/WebEngineering-FHNW/Kolibri/issues/15">
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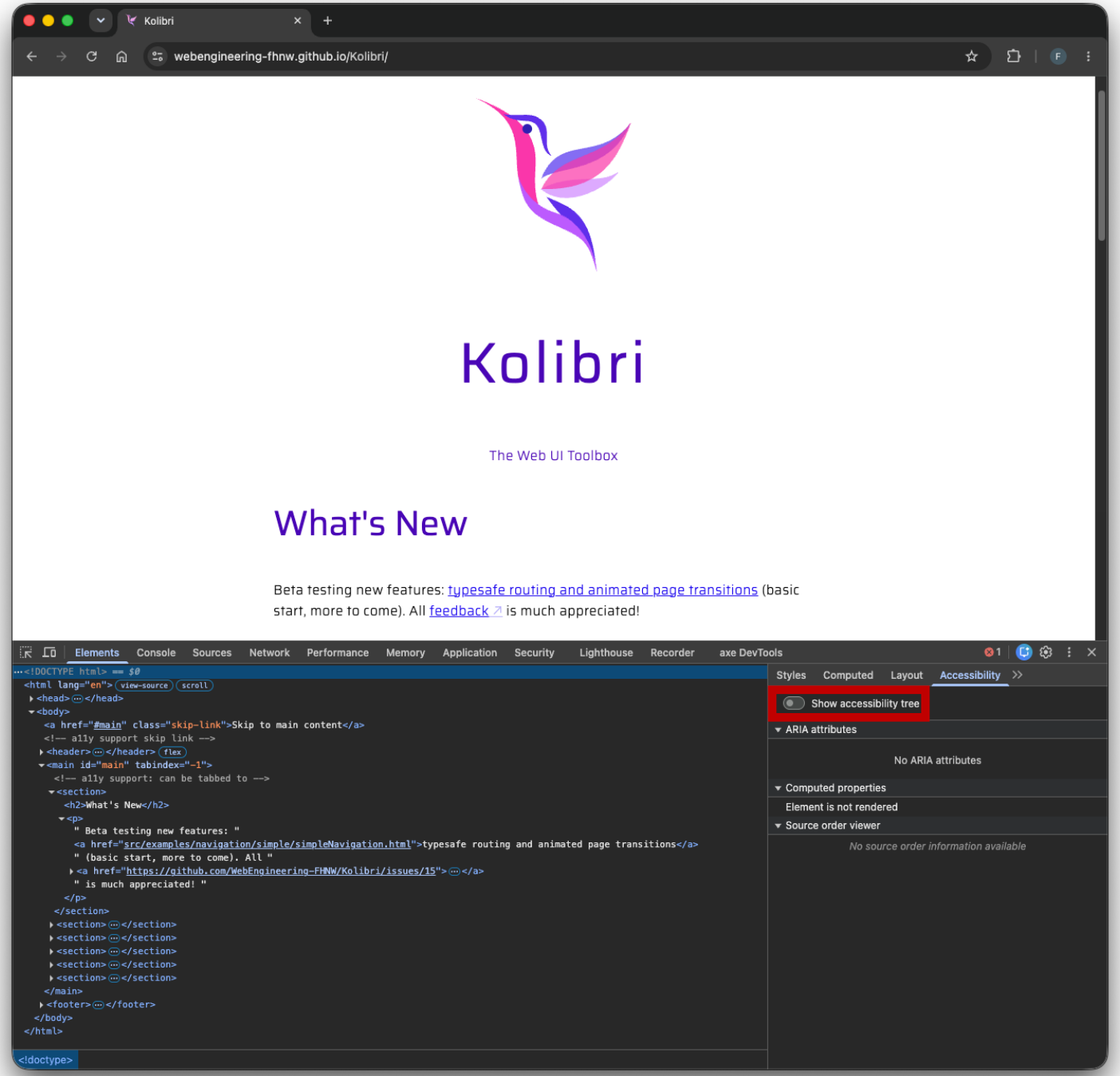
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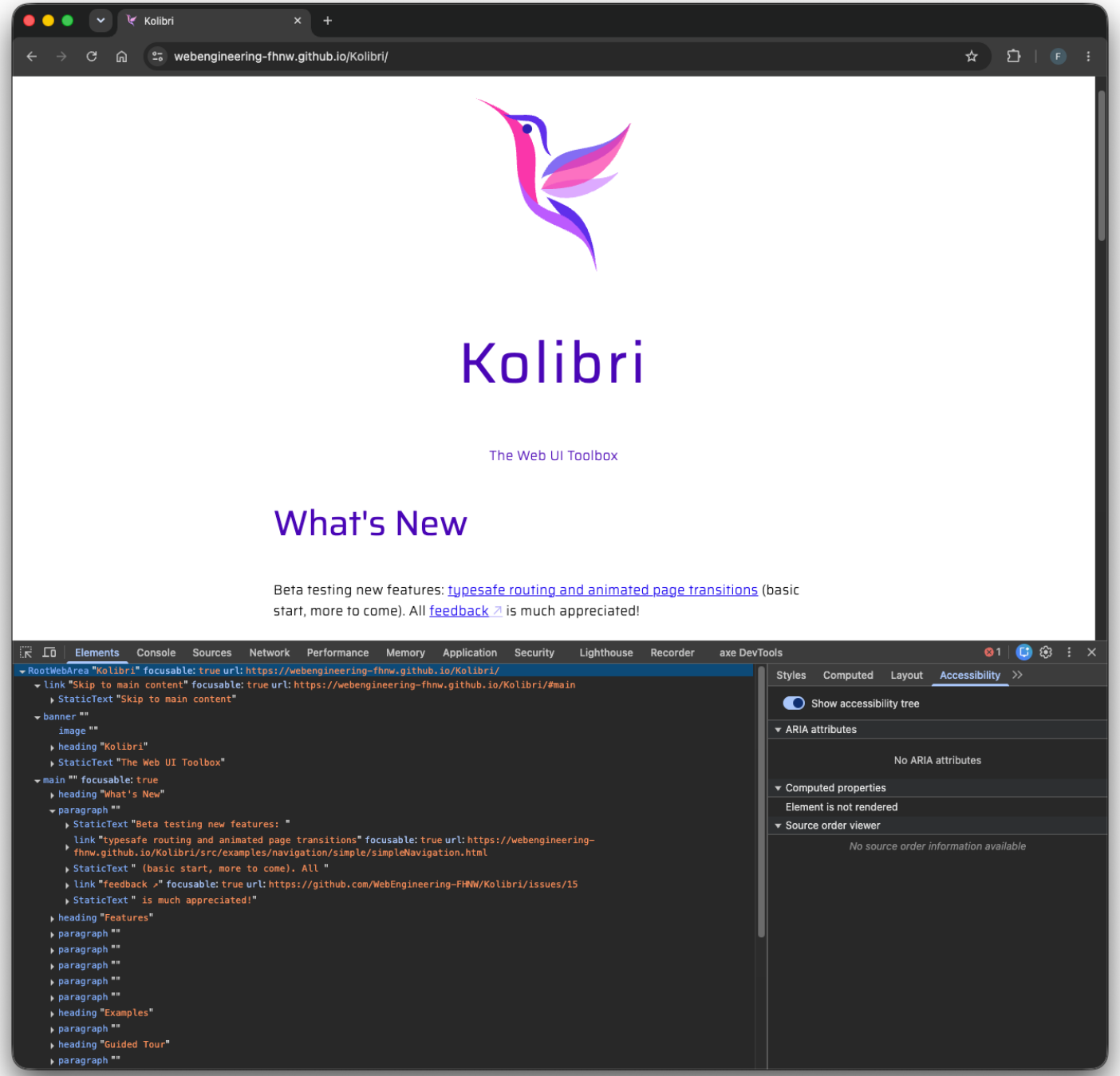
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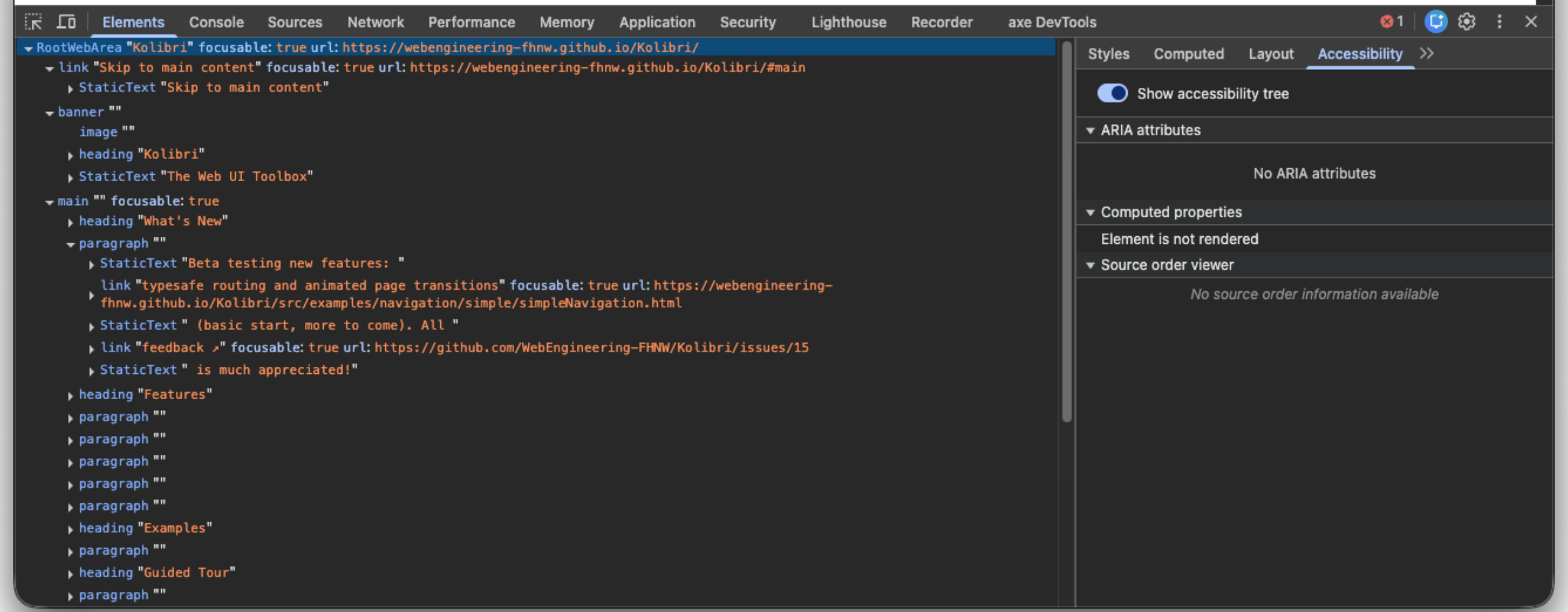
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RootWebArea "Kolibri" focusable: true url: https://webengineering-fhnw.github.io/Kolibri/
link "Skip to main content" focusable: true url: https://webengineering-fhnw.github.io/Kolibri/#main
StaticText "Skip to main content"
banner ""
image ""
heading "Kolibri"
StaticText "The Web UI Toolbox"
main "" focusable: true
heading "What's New"
paragraph ""
StaticText "Beta testing new features: "
link "typesafe routing and animated page transitions" focusable: true url: https://webengineering-fhnw.github.io/Kolibri/src/examples/navigation/simple/simpleNavigation.html
StaticText " (basic start, more to come). All "
link "feedback ↗" focusable: true url: https://github.com/WebEngineering-FHNW/Kolibri/issues/15
StaticText " is much appreciated!"
heading "Features"
paragraph ""
paragraph ""
paragraph ""
paragraph ""
paragraph ""
heading "Examples"
paragraph ""
heading "Guided Tour"
paragraph ""
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```
<button>Button Text</button>
```

```
button "Button Text" focusable: true
  StaticText "Button Text"
    InlineTextBox "Button Text"
```

VS

```
<div>Button Text</div>
```

```
StaticText "Button Text"
  InlineTextBox "Button Text"
```

▶ Berry fun fact

▼ Berry fun fact

Botanically speaking, bananas, watermelons, and pumpkins are all true berries, but strawberries and raspberries aren't.

```
<details name="facts">
  <summary>
    Berry fun fact
  </summary>
  Botanically speaking, bananas, watermelons, and
  pumpkins are all true berries, but strawberries
  and raspberries aren't.
</details>
```

▶ Berry fun fact

▼ Berry fun fact

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```

▶ Berry fun fact

▼ Berry fun fact

Botanically speaking, bananas, watermelons, and pumpkins are all true berries, but strawberries and raspberries aren't.

Uploading Document



```
<label for="progress-bar">
  Uploading Document
</label>
<progress id="progress-bar" value="70" max="100">
  70%
</progress>
```

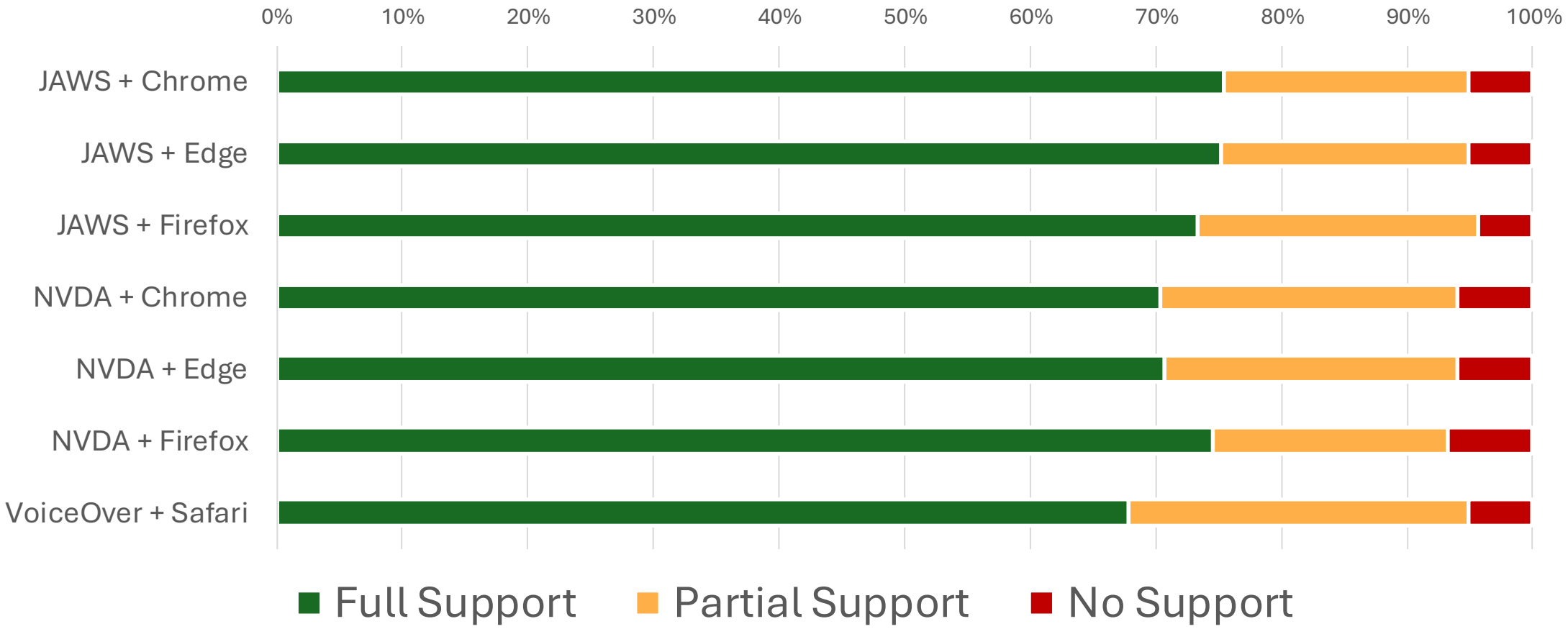


ARIA (Accessible Rich Internet Applications)

First Rule of ARIA Use

"*If you can use a native HTML element or attribute with the semantics and behavior you require* ***already built in****, instead of re-purposing an element and adding an ARIA role, state or property to make it accessible****, then do so***"

ARIA Support



https://a11ysupport.io/

aria-hidden

```
<button type="submit">
  <span class="label">Send Message</span>
  <span class="icon" aria-hidden="true">&#x27A4;</span>
</button>
```



aria-hidden

```
<button type="submit">
  <span class="label">Send Message</span>
  <span class="icon" aria-hidden="true">&#x27A4;</span>
</button>
```



“Send Message Black Rightwards Arrowhead”

aria-label

```
<button type="submit" aria-label="Send message">
  <span class="icon" aria-hidden="true">&#xFE0F;</span>
</button>
```



aria-label

```
<button type="submit" aria-label="Send message">
  <span class="icon" aria-hidden="true">&#xFE0F;</span>
</button>
```



“Send Message”

Bessere Lösung ohne ARIA

```
<button type="submit" class="btn-send"></button>
```

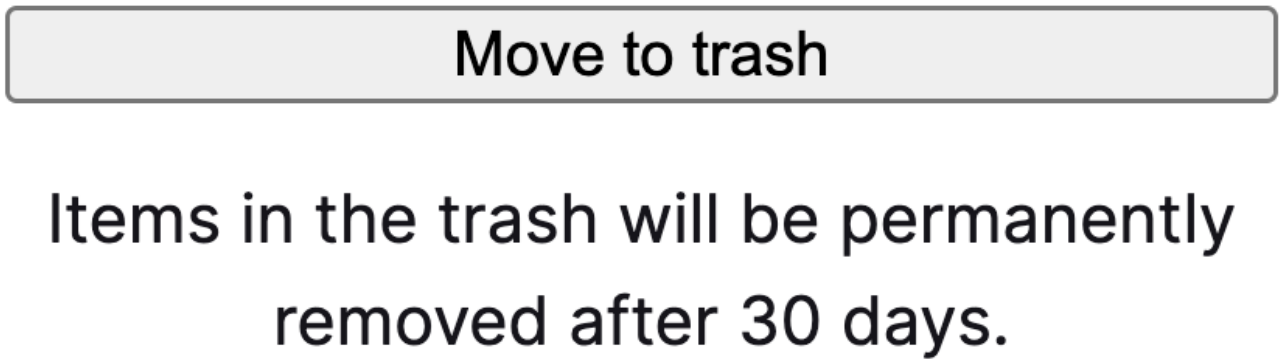
```
.btn-send::before {
  content: "\2709\fe0f" / "Send message";
}
```



“Send Message”

aria-describedby

```
<button aria-describedby="trash-desc">Move to trash</button>
<p id="trash-desc">
  Items in the trash will be permanently
  removed after 30 days.
</p>
```



Semantik in Eigenverantwortung

```
<div
  id="my-button"
  class="button"
  role="button"
  tabindex="0">
  Button Text
</div>
```

Button Text

```css
.button {
  display:          inline-block;
  padding:          2px 7px;
  font:             -webkit-small-control;
  background-color: #EFEFEF;
  border:           1px solid #767676;
  border-radius:    2px;
  cursor:           default;
}

.button:hover {
  background-color: #E5E5E5;
  border-color:     #666666;
}
```

Original vs Fake





```
const button = document.getElementById('my-button');

function handleAction() {
  //Business logic
}

button.addEventListener('click', handleAction);

button.addEventListener('keydown', event => {
  if (event.key === 'Enter' || event.key === ' ') {
    event.preventDefault();
    handleAction();
  }
});
```

WCAG (Web Content Accessibility Guideline)



https://www.w3.org/WAI/WCAG22/wcag2.2AAA-blue.svg

1

Level A (Grundanforderungen)

- Bedeutet eine Seite ist nicht komplett unzugänglich.

2

Level AA (Industrie Standard)

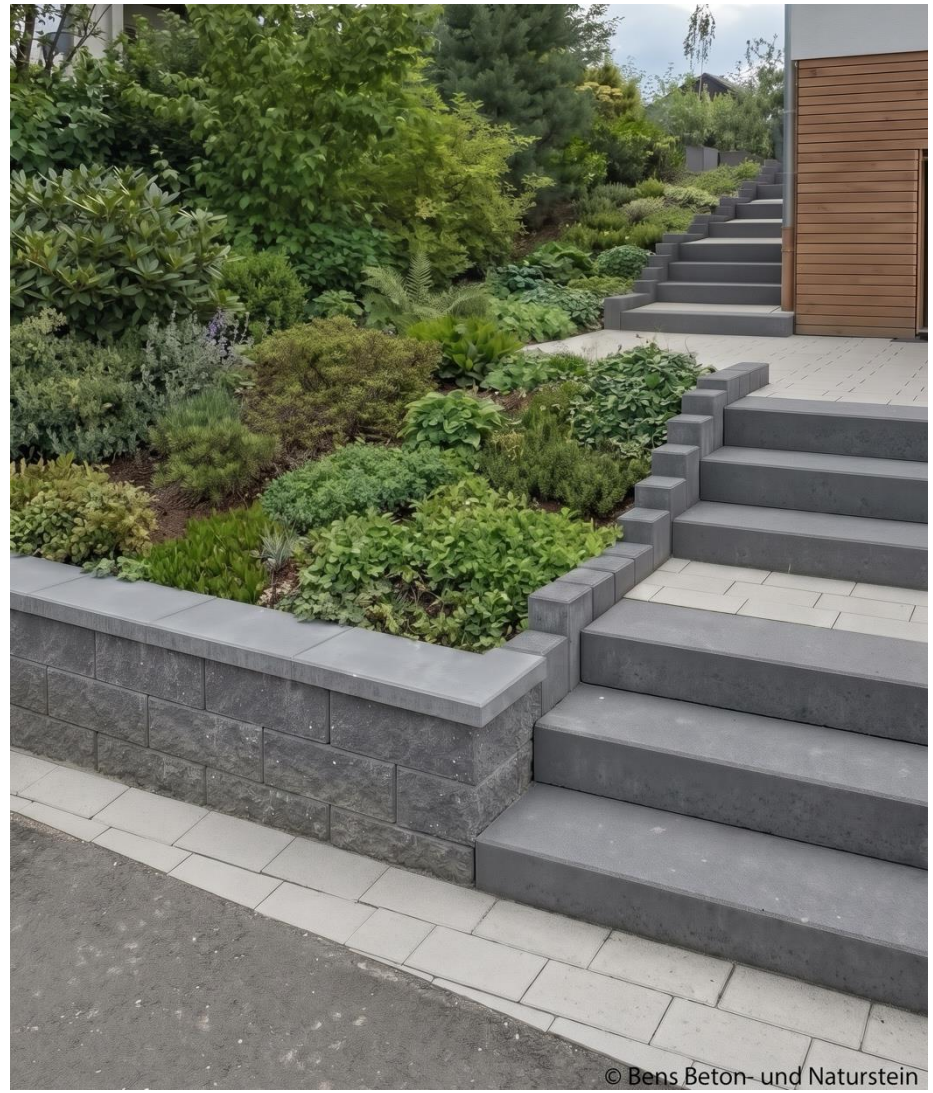
- Standard, auf den sich auch der European Accessibility Act bezieht. (EN 301 549)

3

Level AAA (Gold Standard)

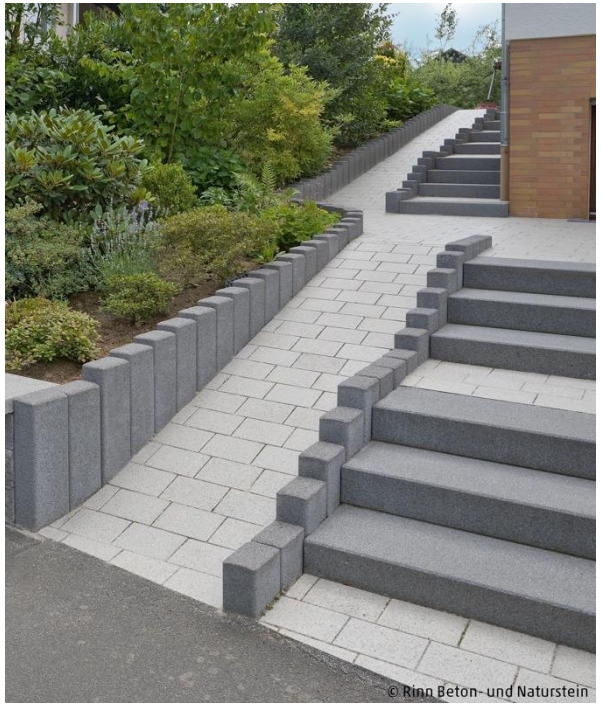
- Kontextbedingt und nicht auf jeden Content anwendbar.

https://www.barrierefreiheit-dienstekonsolidierung.bund.de/Webs/PB/DE/gesetze-und-richtlinien/en301549/en301549-node.html


© Bens Beton- und Naturstein

Level A

Sinnbildliches Alltagsbeispiel



Analoges Web-Beispiel




```
<img
  src="/path/to/lemon.webp"
  alt="Ganze Bio-Zitrone
       mit grünem Blatt" />
```

Success Criterion 1.1.1 Non-text Content

https://i.pinimg.com/originals/f6/a7/83/f6a78356f703f7ecb63e61eec668998e.jpg
https://media01.stockfood.com/largepreviews/NDkxMjUwOA==/00158468-Eine-Zitrone-mit-Blatt.jpg

Level AA

Sinnbildliches Alltagsbeispiel



Analoges Web-Beispiel

Text mit 4.5:1 Kontrast Ratio

```
p {
  background-color: #FFFFFF;
  color:            #BF25E9;
}
```

Success Criterion 1.4.3 Contrast (Minimum)

$$L = 0.2126 \cdot R + 0.7152 \cdot G + 0.0722 \cdot B$$

$$\text{Where } C = \begin{cases} \frac{C_{\text{sRGB}}}{12.92} & \text{if } C_{\text{sRGB}} \leq 0.03928 \\ \left(\frac{C_{\text{sRGB}}+0.055}{1.055}\right)^{2.4} & \text{otherwise} \end{cases}$$

Example: Dodger Blue `rgb(30, 144, 255)`

1. Normalize (/255)

$R_{\text{sRGB}} = \frac{30}{255} = 0.1176$
$G_{\text{sRGB}} = \frac{144}{255} = 0.5647$
$B_{\text{sRGB}} = \frac{255}{255} = 1.0000$

2. Linearize

$R = 0.0123$
$G = 0.2747$
$B = 1.0000$

3. Calculate Luminance (L)

$$L = (0.2126 \cdot 0.0123) + (0.7152 \cdot 0.2747) + (0.0722 \cdot 1.0000) = \mathbf{0.2712}$$

Calculate Contrast Ratio

$$\text{Ratio} = \frac{\text{L1} + 0.05}{\text{L2} + 0.05}$$

$$L = 0.2126 \cdot R + 0.7152 \cdot G + 0.0722 \cdot B$$

$$\text{Where } C = \begin{cases} \frac{C_{\text{sRGB}}}{12.92} & \text{if } C_{\text{sRGB}} \leq \sout{0.03928} \text{ 0.04045} \\ \left(\frac{C_{\text{sRGB}}+0.055}{1.055}\right)^{2.4} & \text{otherwise} \end{cases}$$

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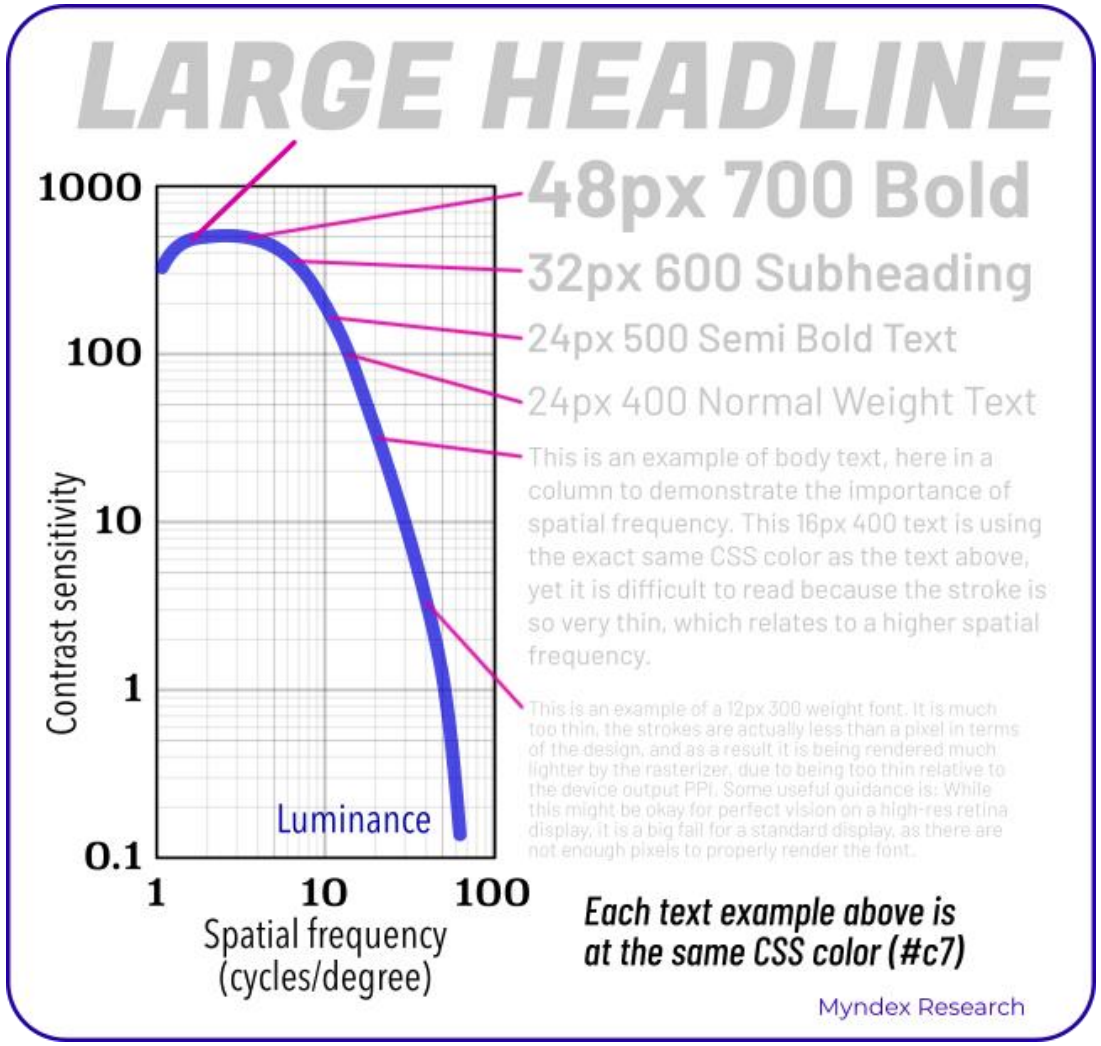
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Calculate Contrast Ratio

$$\text{Ratio} = \frac{\text{L1} + 0.05}{\text{L2} + 0.05}$$

APCA (Accessible Perceptual Contrast Algorithm)



https://git.apcacontrast.com/documentation/WhyAPCA.html

Level AA

Sinnbildliches Alltagsbeispiel



Analoges Web-Beispiel

Text mit 4.5:1 Kontrast Ratio

```
p {
  background-color: #FFFFFF;
  color:            #BF25E9;
}
```

Success Criterion 1.4.3 Contrast (Minimum)

Level AAA

Sinnbildliches Alltagsbeispiel



Analoges Web-Beispiel

Breadcrumbs > showing > **your Location**

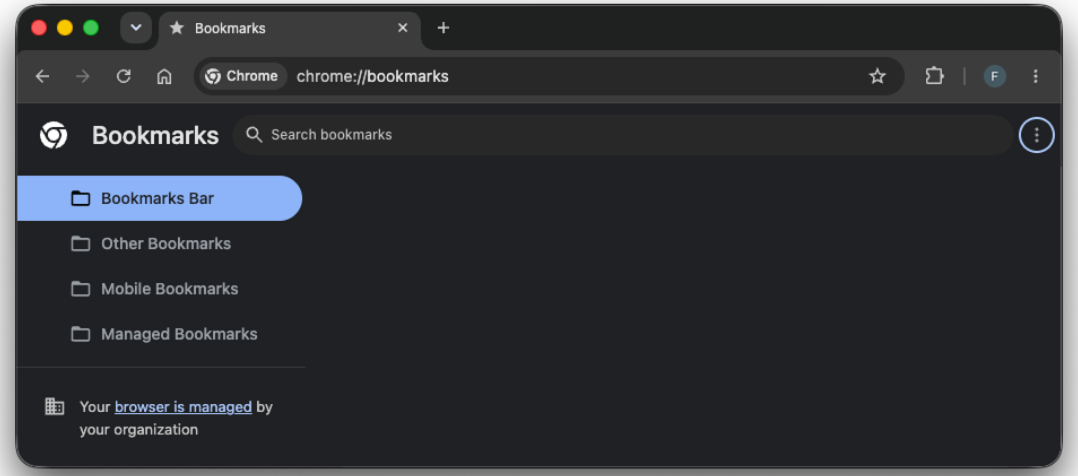
Success Criterion 2.4.8 Location

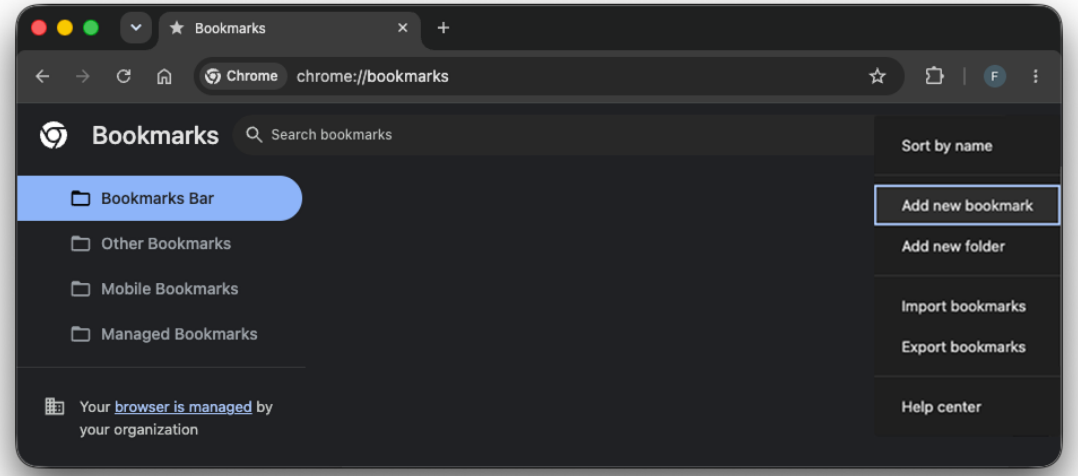
Hands on

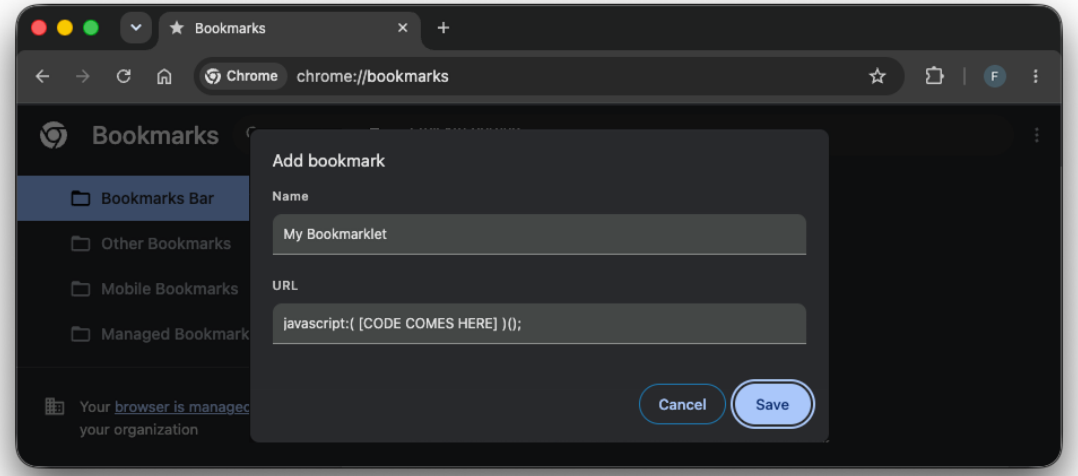
Visual Impairment Übung

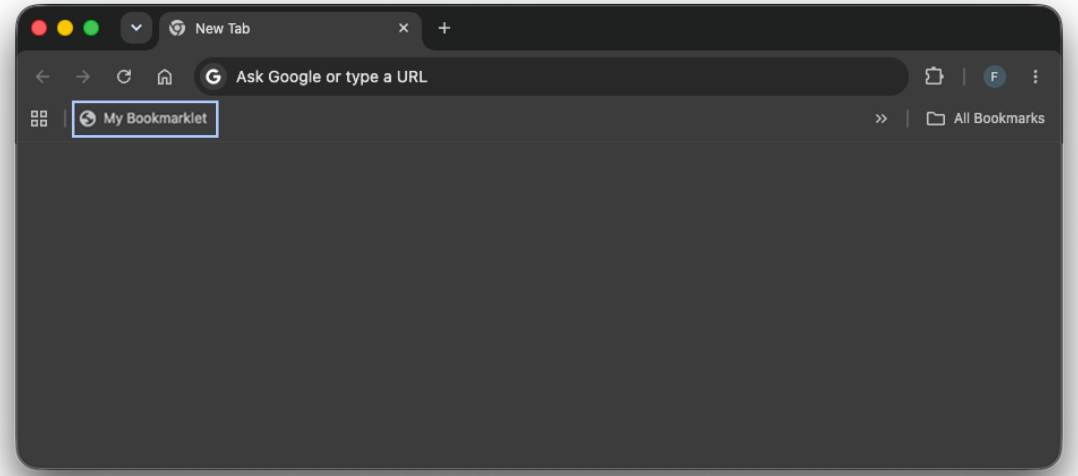
https://github.com/Flaverus/a11y

Bookmarklet - https://github.com/Flaverus/a11y










Kahoot!

Neue Erkentnisse
Anregungen
Themenwünsche

Danke für eure Aufmerksamkeit!